



MONASH
University

ETC3550 **Applied Forecasting**

Final Project

Retail Data Analysis

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BUSINESS
SCHOOL

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1 Visual analysis

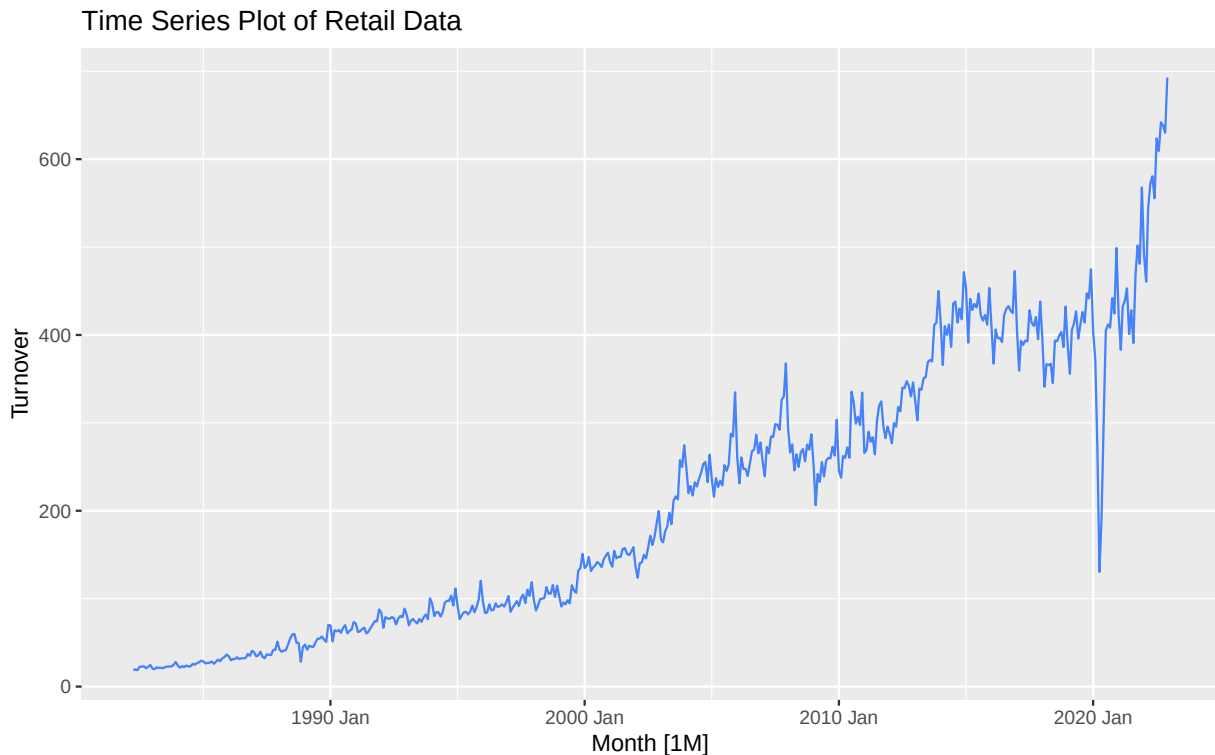


Figure 1

Looking at the plot we can notice a somewhat exponential positive trend.

Additionally, we can note some multiplicative seasonal patterns.

We can also see some indications of cyclical patterns where there appears to be a drop in the late 1980's. Additionally, we can see what appears to be a drop in the late 2000's (likely around the 2008 GFC) and another fall in the early 2020's (likely due to Covid).

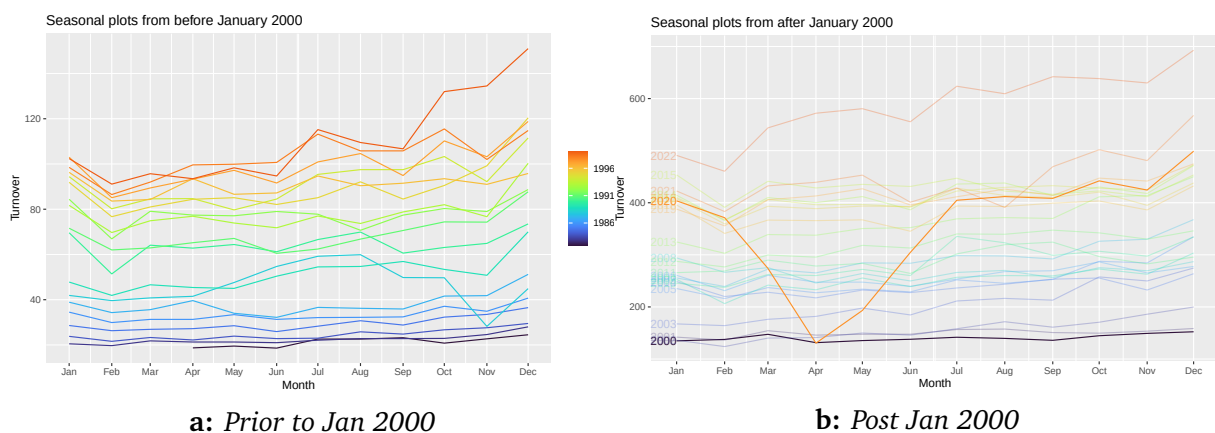


Figure 2: Seasonal Plots

By plotting the data with the `gg_season` function we can note some features of the data.

Specifically, we are able to notice that as the years increment, so does the data increase (indicating a positive trend). Furthermore, we can notice how there is an increase in sales by December and January, however, there is then a drop in sales during the month of February.

Additionally, we can note that there are some years with a significant drop in sales. Looking the first plot we can see a significant drop in the late 1980's (Green line) around the end of the year. Furthermore, in the second plot we can see another significant dip in 2020 (Purple line) which is also around the time of the covid outbreak.

Nevertheless, there doesn't appear to be any significant drops in the late 2000's despite what it looked like in the autoplot plot.

Ultimately, these drops and recoveries are part of the cyclical behaviour of this data.

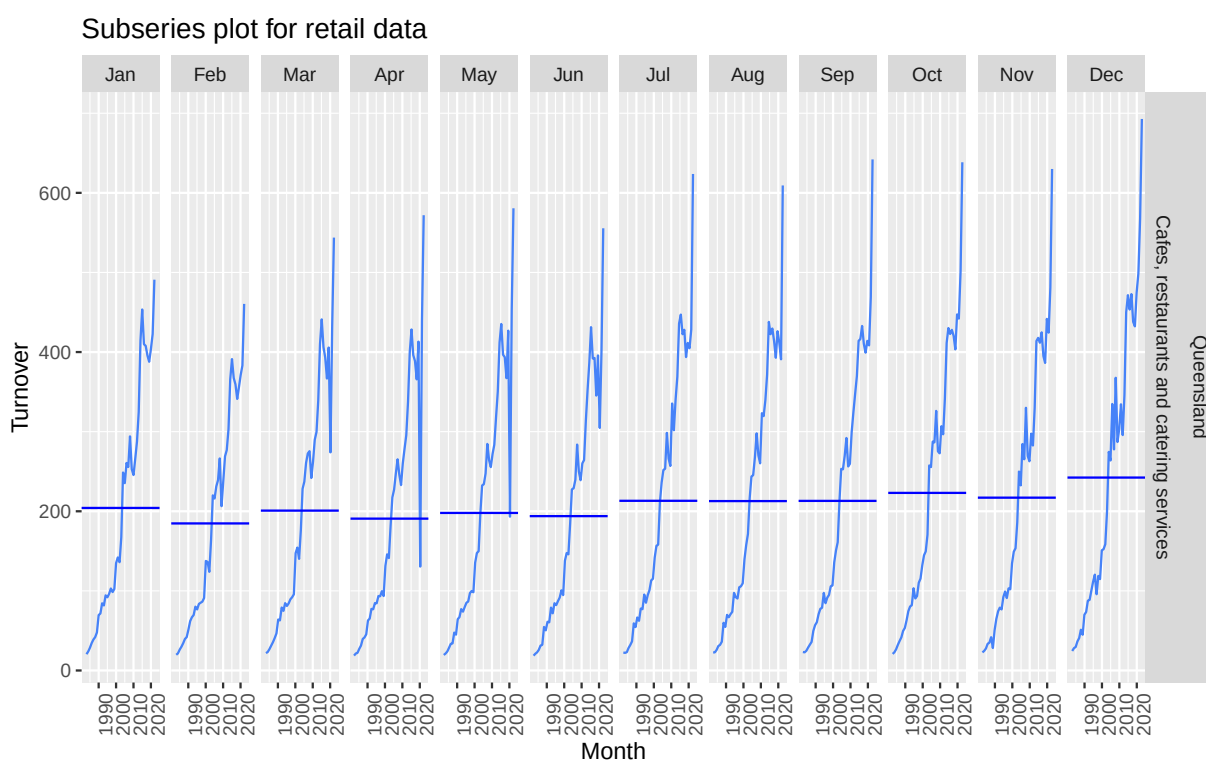


Figure 3

Taking a look at the `gg_subseries` plot we can note some features which were hidden in the other 2 plots.

First, we can notice how there is a dip in the late 2000's over every month which is possibly the reason it wasn't so obvious in the `gg_season` plot.

Furthermore, we are also able to notice how the drop due to covid is extreme during the months of March to June. However, after June the drop seems minimal compared to the earlier months.

2 Statistical features and effects of Covid

Looking at Figure 1 we can note an exponentially increasing trend and multiplicative seasonality and we can clearly see a large dip that quickly in early 2020, which is due to the effects of Covid.

Furthermore, we can learn more about the effects of Covid from Figure 2b where we can see how severe, but short lived the effect of Covid was. We can easily notice that the turnover in 2020 drops to the same turnover as the year 2000 in April, although the dip was short lived and recovered to pre Covid levels by July 2020.

3 Box-Cox Transformation

To find a good λ value for the Box Cox transformation, we will first plot the data with $\lambda = 0$ (log transformation) and $\lambda = 1$ (original data)

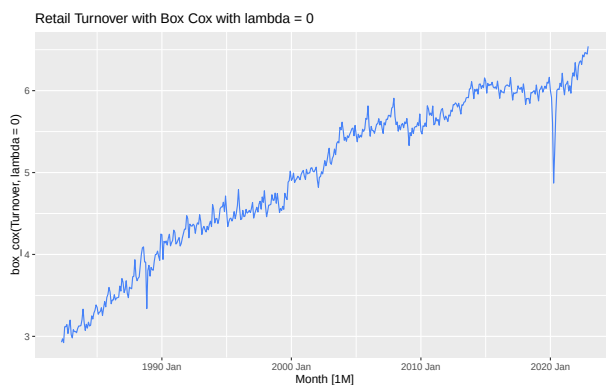


Figure 4: $\lambda = 0$

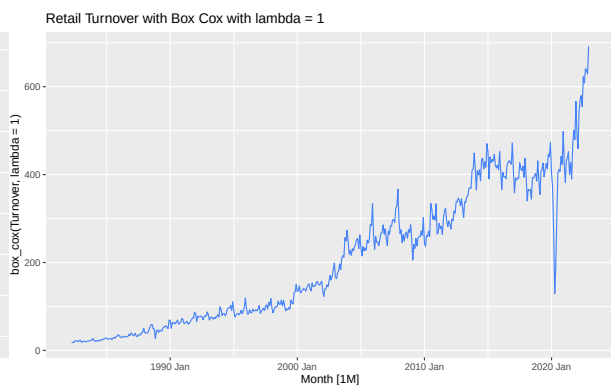


Figure 5: $\lambda = 1$

Firstly, we are able to see that the variation clearly looks multiplicative with the identity transformation. Furthermore, We can see here that the $\lambda = 0$ transformation is pretty good. however, there is a bit of variation in the seasonality in the middle of the data period.

Now we will use the Guerrero feature and see if it beats the log function.

Retail Turnover with Box Cox with $\lambda = -0.04$ using Guerrero

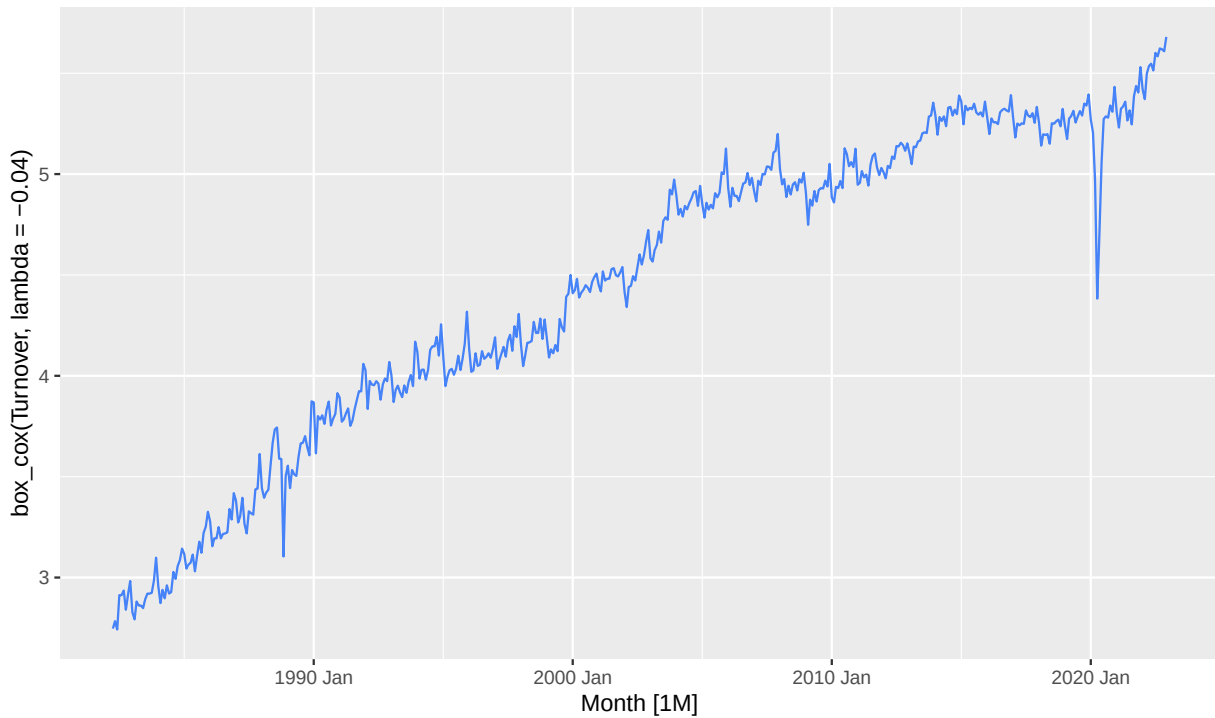


Figure 6

We can see here that the Guerrero feature doesn't improve on $\lambda = 0$, thus we will just use $\lambda = 0$ as it is the simple log transformation.

4 STL Decomposition

We can now decompose the transformed data using STL decomposition.

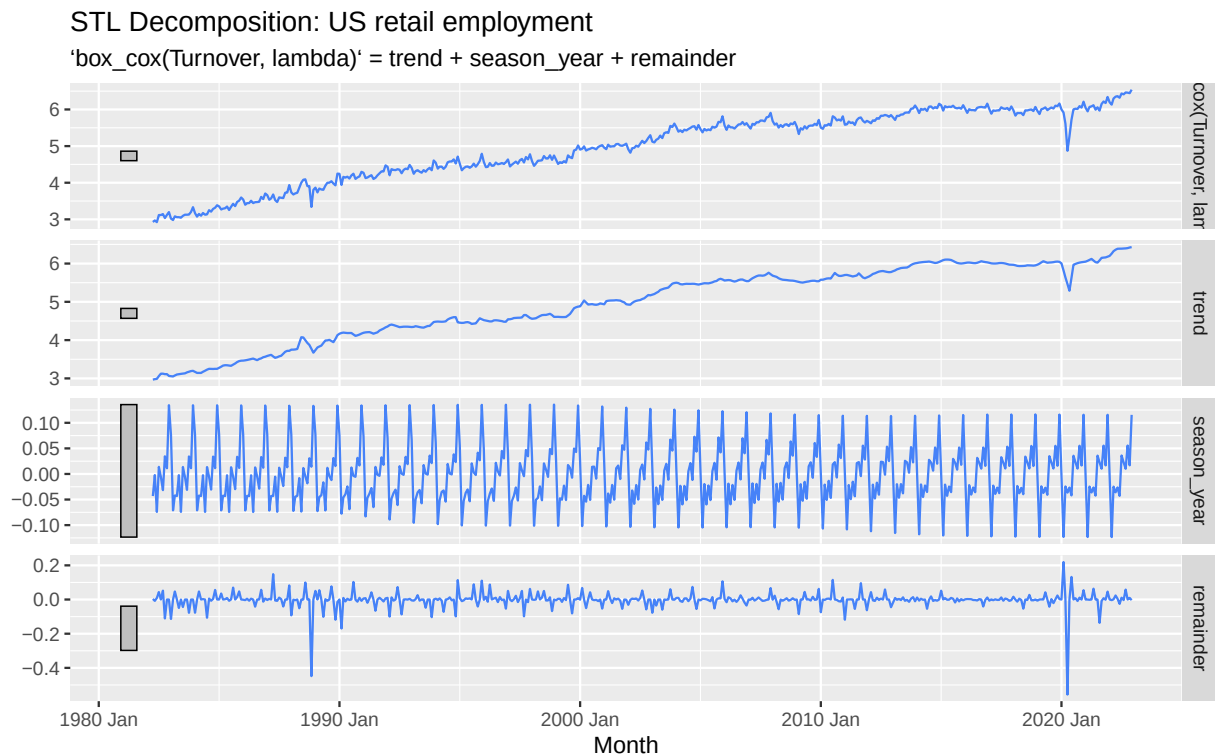


Figure 7

While some seasonality remains in the remainder, we can see some regular features of the data, such as positive trend and some cyclical patterns.

There is one particular feature which stands out in the STL Decomposition, that is how the seasonality changes as the years progress. As in up until around 2010, we can see that the December/January increase in sales was much more significant than since 2010. This can be seen by how the peaks seem to be “fatter” in the later years, whereas, they are a lot narrower in the earlier years. This may be due to internet shopping which helps consumers make easy purchases at all times of the year.

The choice of window parameters were selected mainly in attempt to reduce the remainder values, whilst trying to balance the smoothness of the trend and reduce the variations between seasons.

5 Differencing to make data stationary

Now if we again look at Figure 1, we can see that there is clear season and trend, thus we will first take a first order difference.

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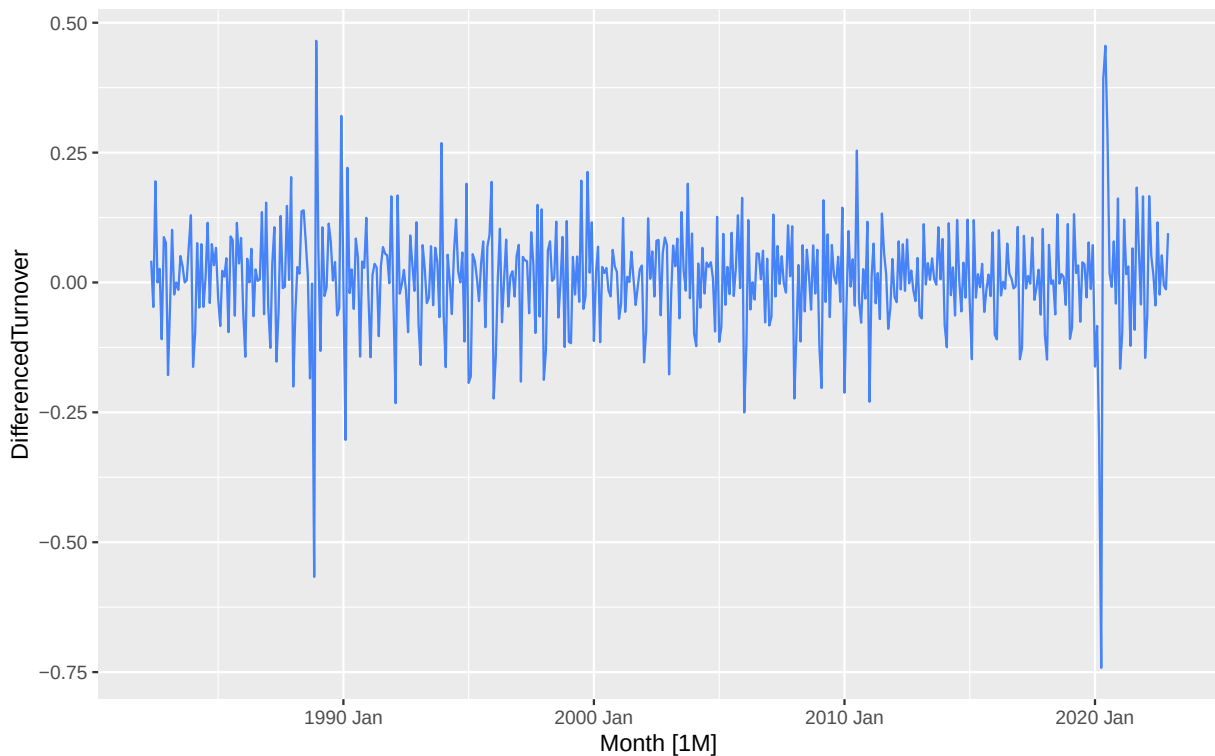


Figure 8

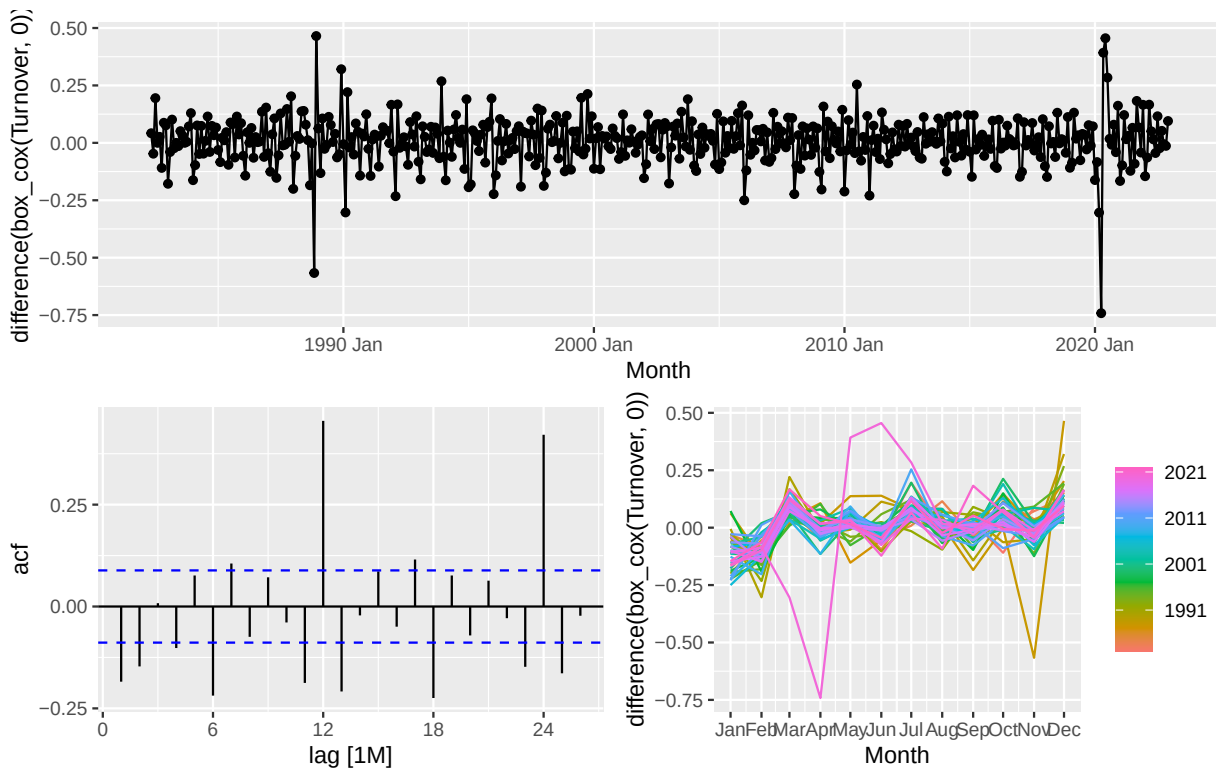
This data, looks like it could possibly be stationary, as it is roughly horizontal and has mostly constant variance as well as there are no obvious signs of seasonal patterns. However, we will use the KPS test to determine if we need more than a first order difference.

State	Industry	kpss_stat	kpss_pvalue
Queensland	Cafes, restaurants and catering services	0.0496331	0.1

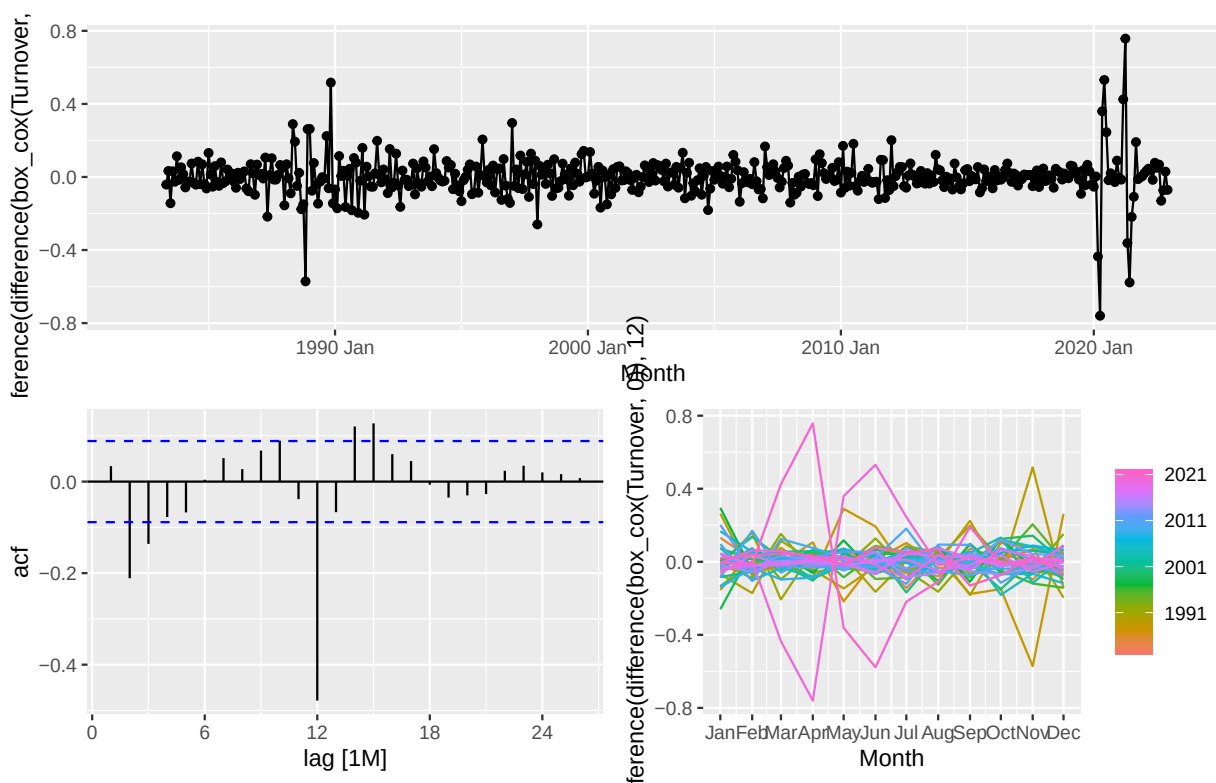
Here we see that $p > 0.05$ thus there is insufficient evidence to reject the null, meaning that the data is likely stationary. Thus, we don't require another order of differencing'.

Now if we look at the ACF plot of the data we can see there should be any more differencing.

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Here we can clearly see some spikes in the seasonal lags, thus a seasonal difference should be taken as well.



Now, the ACF plot looks better and we can do another unit root test just to ensure it is stationary.

State	Industry	kpss_stat	kpss_pvalue
Queensland	Cafes, restaurants and catering services	0.0103342	0.1

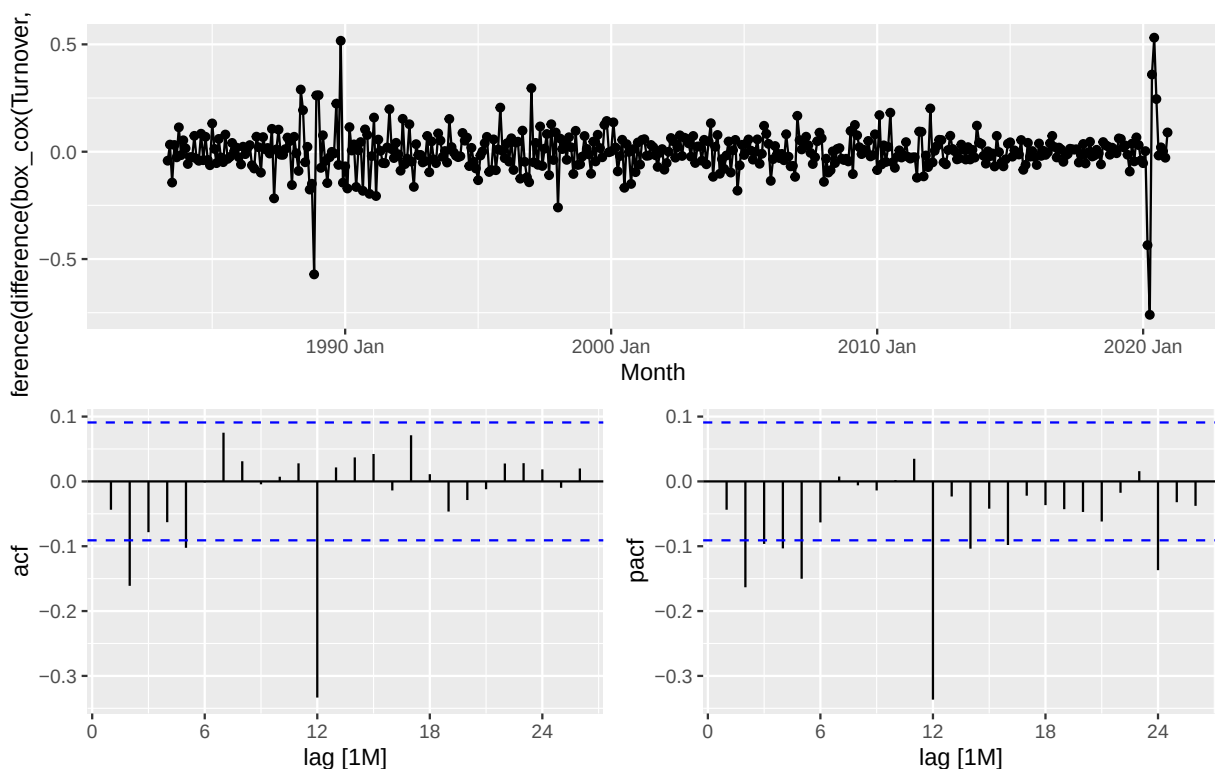
And since $p > 0.05$ we can still say that the data is stationary.

6 Short list of ETS and ARIMA models

Now we can look at Figure 1 and note that there is a multiplicative trend and multiplicative seasonality, thus we will fit an ETS(M,A,M).

Additionally, we will fit an ETS(A,A,A) on the log transformed turnover data. Furthermore, we will fit the default parameters for both the original data and transformed data.

Now we will also look at ARIMA models by first looking at the ACF/PACF plots



Looking at the ACF/PACF plots we can use an ARIMA(0,1,2)(0,1,1)(12) model or alternatively, ARIMA(5,1,0)(2,1,0)(12) model for our data. Additionally, we will use grid search to find an alternative specification. Furthermore, since there are 2 orders of differencing, we will not use a constant in our model.

Now we can look at the AICc values of each of the models, however, we can't use these values to determine the best one as they have different coefficients and different orders of differencing.

Model name	AICc
ets_MAM	5052.4881
ets_default	5047.2168
ets_transformed_AAA	539.7294
ets_transformed_default	539.7294
arima_510_210	-927.0185
arima_012_011	-981.6568
arima_search	-986.4772

We can look at the RMSE, however, which we can use to compare the various models

.model	Specification	RMSE	AICc	AIC	ME	MAE.x	MPE	MAPE	MASE	RMSSE
ets_transformed_AAA	<ETS(A,A,A)>	72.1970	539.7294	538.3604	1.8338	57.1685	52.4804	50.2860	67.7792	24238299
ets_transformed_default	<ETS(A,A,A)>	72.1970	539.7294	538.3604	1.8338	57.1685	52.4804	50.2860	67.7792	24238299
arima_012_011	<ARIMA(0,1,2)(0,1,1)>	19.7124	981.6568	981.7463	72.8415	80.8625	82.3796	94.3729	13070807	37092026
ets_default	<ETS(M,Ad,M)>	119.0045	5047.2168	5045.6832	262439	44.7080	24.8181	18.5820	667678	38689448
ets_MAM	<ETS(M,A,M)>	119.0540	5052.4881	5051.1191	114965	35.4963	15.6620	7.7554	880750	32690993
arima_510_210	<ARIMA(5,1,0)(2,1,0)>	927.0185	927.3435	927.3435	99.4821	106.1454	142.7298	154.4006	5234364	3174304
arima_search	<ARIMA(1,1,2)(2,0,0)> w/ drift	127.6504	986.4772	986.7228	127.6504	132.1842	121.7235	134.8652	2065443	40965140

Looking at the RMSE, we can see that the ETS(A,A,A) on the transformed series performs the best. Additionally, we can see that the ARIMA(0,1,2)(0,1,1)[12] model is the best performing arima model.

7 Analysis of best ETS and ARIMA models

Now that we found that ETS(A,A,A) and ARIMA(0,1,2)(0,1,1)[12] perform the best on the test set, we will use these models.

7.1 Parameter estimates

First we will show the parameter estimates.

.model	term	estimate	std.error	statistic	p.value
ets_transformed_AAA	alpha	0.8954923	NA	NA	NA
ets_transformed_AAA	beta	0.0163615	NA	NA	NA
ets_transformed_AAA	gamma	0.0001003	NA	NA	NA
ets_transformed_AAA	l[0]	2.9644098	NA	NA	NA
ets_transformed_AAA	b[0]	0.0152919	NA	NA	NA
ets_transformed_AAA	s[0]	-0.0315271	NA	NA	NA
ets_transformed_AAA	s[-1]	-0.0999976	NA	NA	NA
ets_transformed_AAA	s[-2]	0.0180956	NA	NA	NA
ets_transformed_AAA	s[-3]	0.1349630	NA	NA	NA
ets_transformed_AAA	s[-4]	0.0145510	NA	NA	NA
ets_transformed_AAA	s[-5]	0.0542093	NA	NA	NA
ets_transformed_AAA	s[-6]	0.0066626	NA	NA	NA
ets_transformed_AAA	s[-7]	0.0311957	NA	NA	NA
ets_transformed_AAA	s[-8]	0.0287650	NA	NA	NA
ets_transformed_AAA	s[-9]	-0.0548434	NA	NA	NA
ets_transformed_AAA	s[-10]	-0.0372857	NA	NA	NA
ets_transformed_AAA	s[-11]	-0.0647885	NA	NA	NA
arima_012_011	ma1	-0.1462161	0.0485787	-3.009882	0.0027597
arima_012_011	ma2	-0.3005472	0.0527525	-5.697311	0.0000000
arima_012_011	sma1	-0.9084029	0.0329069	-27.605201	0.0000000

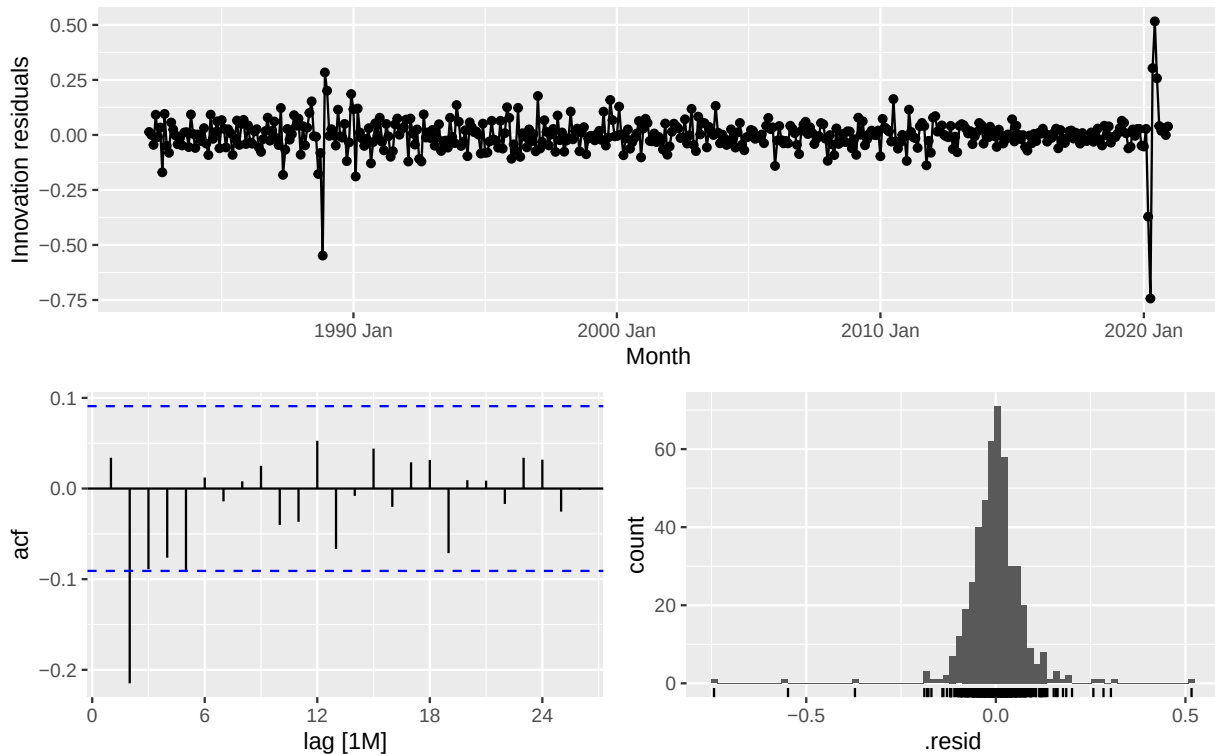
And now we can check the residuals of both models.

7.2 Residual diagnostics

7.2.1 ETS(A,A,A) residual analysis

First, we will look at the residuals of the ETS(A,A,A) model.

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Looking at the innovations, we can see that there are 2 spikes in the residuals, however, the residuals have constant variance other than that. Furthermore, there is a significant spike in the ACF plot which may indicate that there is some autocorrelation which is uncaptured.

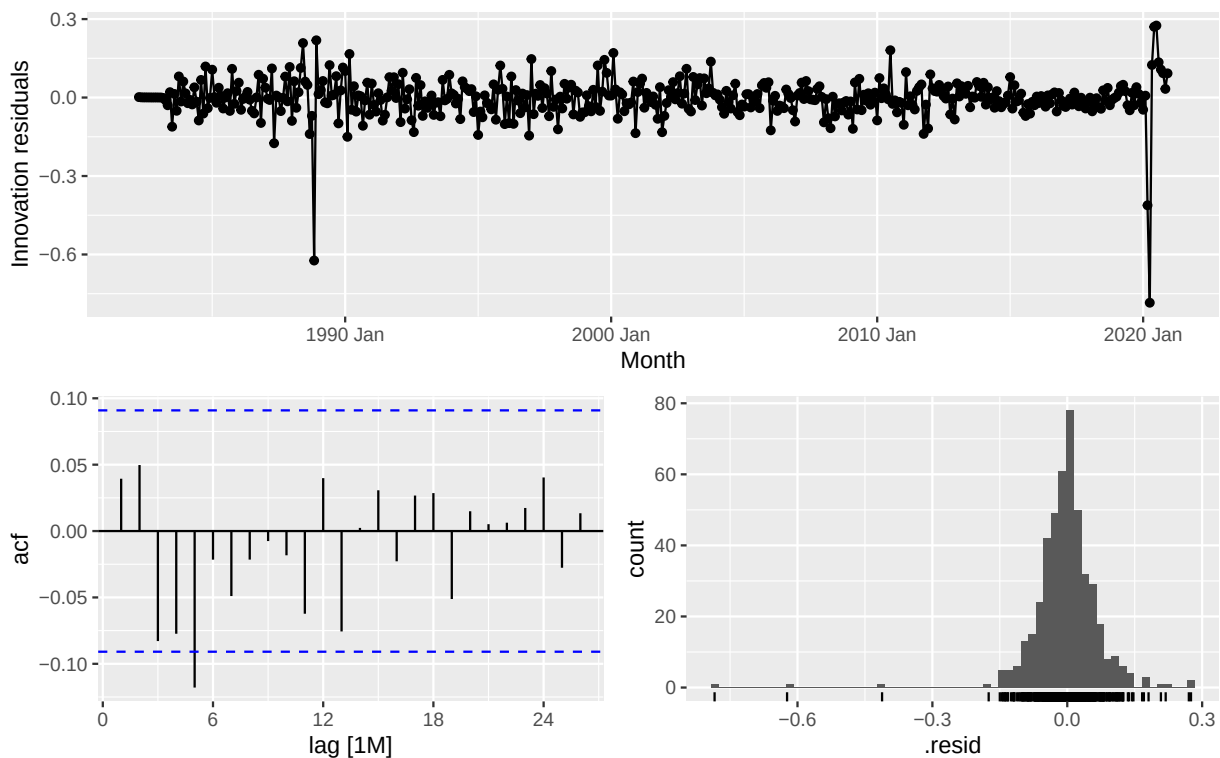
However, looking at the histogram, the residuals look normal.

Nevertheless, we will use the Ljung-Box test to check if the residuals can be considered as white noise

```
[1] 0.0001946861
```

The Ljung Box test finds a p-value less than 0.05 which means that we have sufficient evidence to reject the null which implies that the innovations aren't white noise.

7.2.2 ARIMA(0,1,2)(0,1,1)[12] residual analysis



Looking at the innovations, we can see that there are 2 spikes in the residuals, however, the residuals have constant variance other than that. Furthermore, there is a significant spike in the ACF plot which may indicate that there is some autocorrelation which is uncaptured, however, this spike isn't too large.

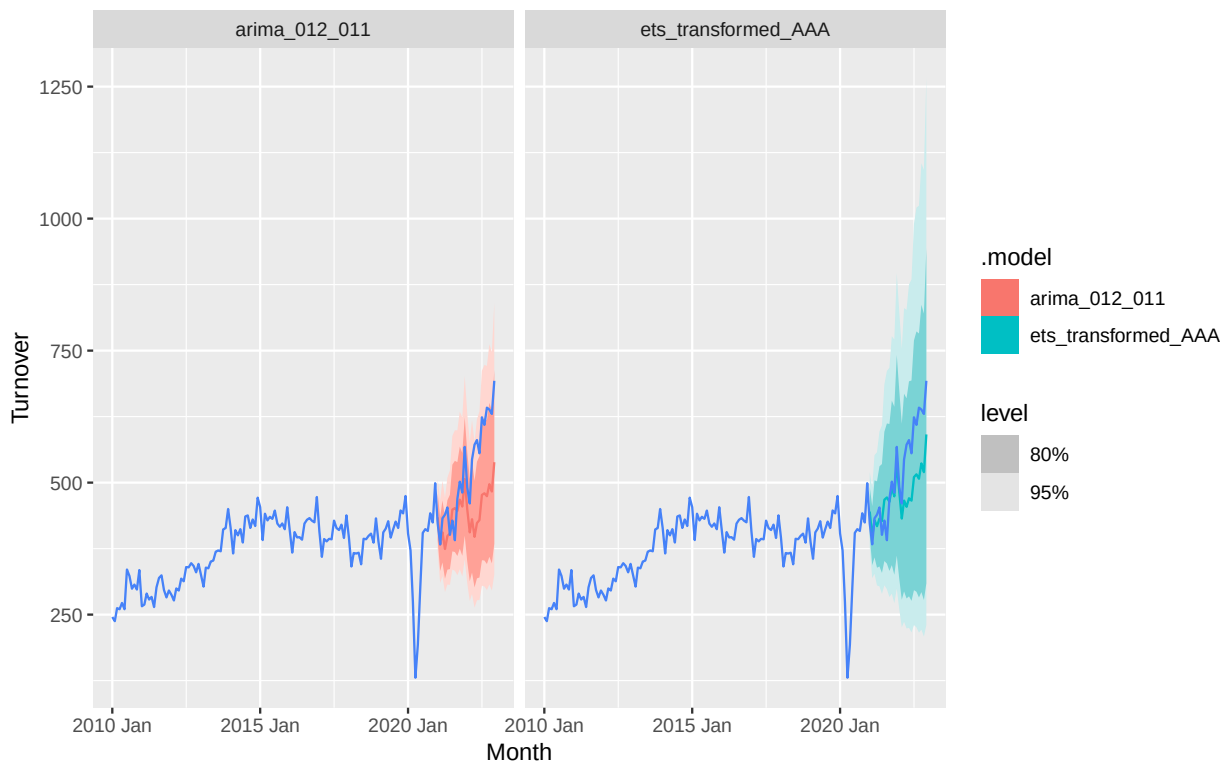
However, looking at the histogram, the residuals look normal.

Nevertheless, we will use the Ljung-Box test to check if the residuals can be considered as white noise

```
[1] 0.09256388
```

The Ljung Box test confirms that the residuals are white noise with a p-value greater than 0.05 which means that we have insufficient evidence to reject the null which implies that the innovations are likely white noise.

7.3 Forecasts and prediction intervals

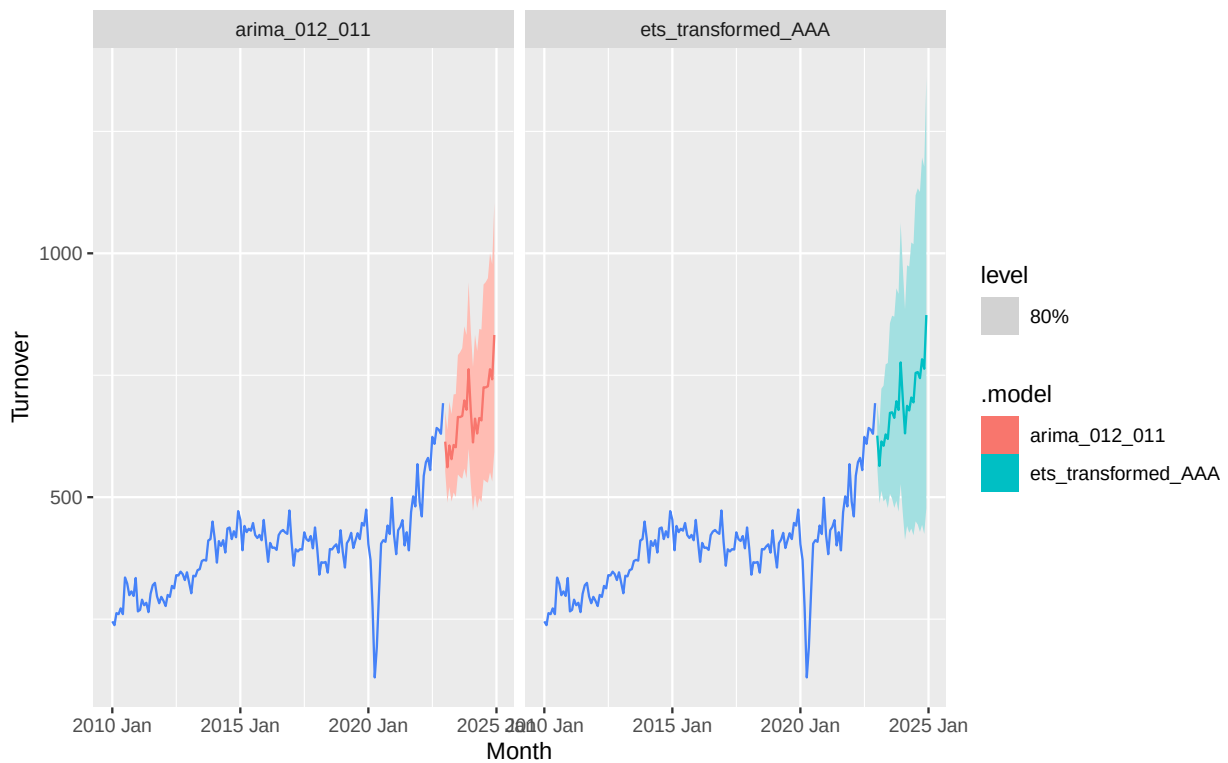


Looking at the 2 plots, we can clearly see that the prediction intervals for the ETS model are much larger, which is likely to do with the data being transformed. However, while the intervals for the ARIMA model are smaller, they still mostly contain the test data, particularly the 95% intervals.

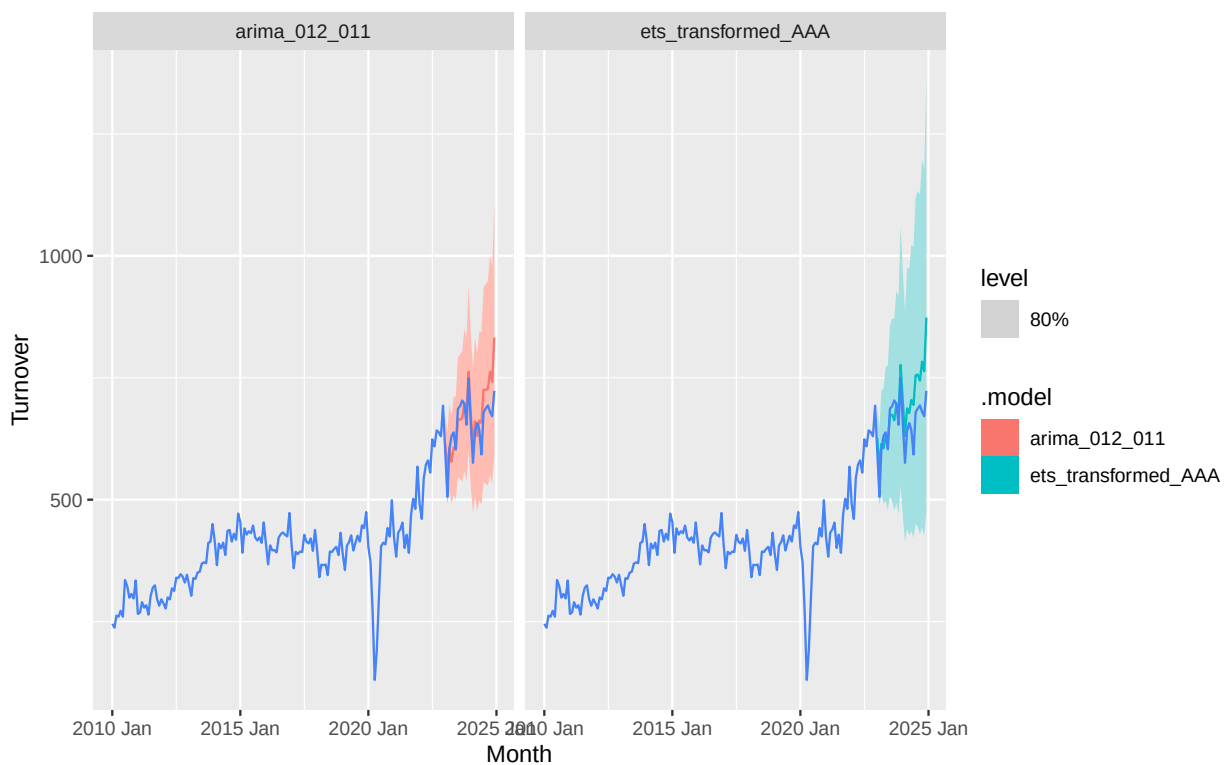
8 Model with better forecasts

Whilst the ETS model doesn't have normal residuals and wider prediction intervals than the ARIMA model, nevertheless, the RMSE for the ETS model is smaller than the ARIMA model and the test data remains in the prediction intervals.

9 Out of sample forecasts



10 Comparison with up to date data



Looking at the data, we can see that the new data lies well within the 80% prediction intervals,

therefore, we will look at the RMSE to see which forecasting model performed better.

```
# A tibble: 2 x 8
```

	.model	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	arima_012_011	-19.1	44.0	35.6	-2.98	5.47	NaN	NaN
2	ets_transformed_AAA	-38.7	59.6	47.6	-6.00	7.31	NaN	NaN

Looking at the RMSE, we can see that the arima model outperformed the ETS model.

11 Benefits and limitations of the selected models

A benefit of the ARIMA model is tighter prediction intervals which allows us to make more certain predictions, a disadvantage is that it requires the data to be stationary, which if we don't keep checking the data, might require a different order of differencing as time progresses.

The benefits of the ETS model is that it allows for complex trends and seasonality, which is an advantage over ARIMA models. However, transforming the data forces larger prediction intervals which like said earlier is harder to use for decision making.